

Career Progression and Promotion Gap Analysis for Retention Optimization at Palo Alto Networks

Data Science & HR Analytics Project

1.Tools used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Power BI
- Streamlit

2. Abstract

This project focuses on analyzing employee career progression and identifying promotion gaps that may contribute to career stagnation and retention challenges. The analysis uses employee-level HR data containing demographic, job, compensation, satisfaction, training, and career progression information.

The project applies exploratory data analysis, feature engineering, and unsupervised machine learning techniques to identify different career progression patterns. Metrics such as Promotion Gap Ratio, Role Stagnation Index, Training Intensity, and Manager Stability Ratio are developed to provide additional insight into employee career trajectories.

K-Means clustering is used to group employees according to their career progression characteristics. A Promotion Gap Score is then developed to classify employees into low, medium, and high promotion-gap categories. Finally, potential retention opportunities are identified by combining career stagnation indicators with employee retention-related characteristics.

The results are presented through Power BI and Streamlit dashboards to provide an interactive view of career progression, promotion gaps, training needs, and potential retention opportunities.

3.Introduction

Employee retention is an important organizational challenge because employees may experience career stagnation even when they are not currently disengaged from their organization. Long periods without promotion, extended time in the same role, limited training opportunities, and changes in managerial relationships can provide useful signals about career progression.

Traditional employee attrition analysis primarily focuses on identifying employees who have already left or may leave. However, such analysis may not fully explain the structural career factors associated with employee progression.

This project therefore focuses on career progression intelligence. Instead of concentrating only on attrition prediction, the analysis examines promotion gaps, role stagnation, training participation, manager continuity, and career trajectory patterns.

4. Problem Statement

Organizations require a data-driven approach to understand employee career progression and identify potential promotion stagnation. Without such analysis, HR teams may have limited visibility into employees who remain with the organization but experience extended periods without career advancement.

The objective of this project is to analyze employee career trajectories, identify promotion gaps, classify employees according to career progression patterns, and identify potential retention opportunities.

5. Objectives

1. Analyze employee career progression patterns.
2. Identify employees experiencing long promotion gaps.
3. Measure role stagnation using career-related variables.
4. Analyze training participation and development opportunities.
5. Examine manager continuity and its relationship with career progression.
6. Apply clustering to identify different career trajectory groups.
7. Develop a Promotion Gap Score.
8. Identify potential retention opportunities.
9. Build interactive Power BI and Streamlit dashboards.
10. Provide data-driven insights for HR decision-making.

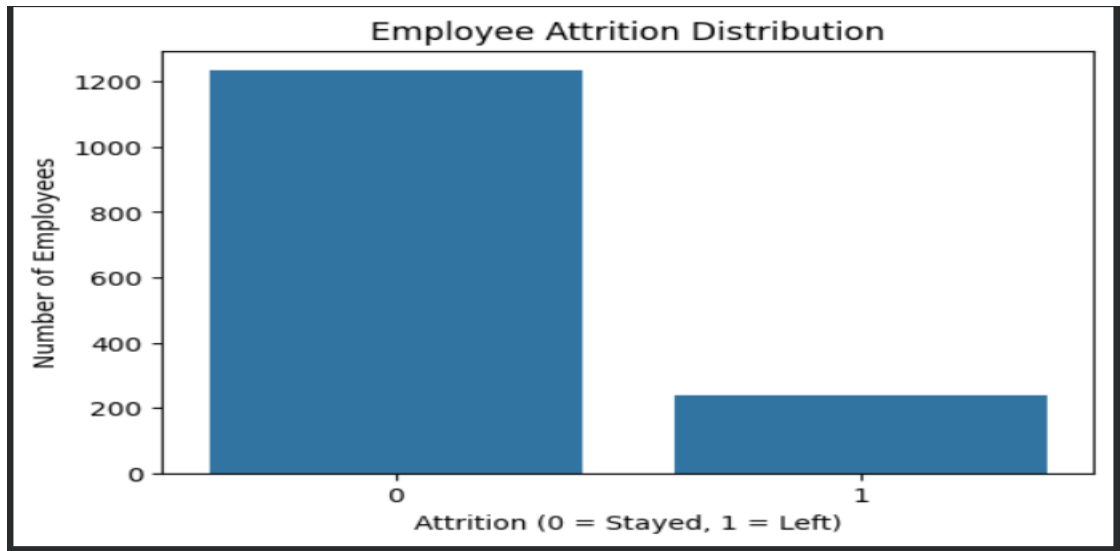
6. Dataset Description

The dataset contains **1,470 employee records and 31 original attributes**. The variables include employee demographics, job characteristics, compensation, satisfaction, training, and career progression information.

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2	41	1	Travel_Rare	1102	Sales	1	2	Life Science	2	Female	94	3	2	Sales Executive	4	Single	5993	19479	8	Yes	11	3	1						
3	49	0	Travel_Frequently	279	Research & Development	8	1	Life Science	3	Male	61	2	2	Research Scientist	2	Married	5130	24907	1	No	23	4	4						
4	37	1	Travel_Rare	1373	Research & Development	2	2	Other	4	Male	92	2	1	Laboratory Technician	3	Single	2090	2396	6	Yes	15	3	2						
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6	27	0	Travel_Rare	591	Research & Development	2	1	Medical	1	Male	40	3	1	Laboratory Technician	2	Married	3468	16632	9	No	12	3	4						
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12	35	0	Travel_Rare	809	Research & Development	16	3	Medical	1	Male	84	4	1	Laboratory Technician	2	Married	2426	16479	0	No	13	3	3						
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14	31	0	Travel_Rare	670	Research & Development	26	1	Life Science	1	Male	31	3	1	Research Scientist	3	Divorced	2911	15170	1	No	17	3	4						
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19	22	0	Non-Travel	1123	Research & Development	16	2	Medical	4	Male	96	4	1	Laboratory Technician	4	Divorced	2935	7324	1	Yes	13	3	2						
20	53	0	Travel_Rare	1219	Sales	2	4	Life Science	1	Female	78	2	4	Manager	4	Married	15427	22021	2	No	16	3	3						
21	38	0	Travel_Rare	371	Research & Development	2	3	Life Science	4	Male	45	3	1	Research Scientist	4	Single	3944	4306	5	Yes	11	3	3						
22	24	0	Non-Travel	673	Research & Development	11	2	Other	1	Female	96	4	2	Manufacturing	3	Divorced	4011	8232	0	No	18	3	4						
23	36	1	Travel_Rare	1218	Sales	9	4	Life Science	3	Male	82	2	1	Sales Representative	1	Single	3407	6986	7	No	23	4	2						
24	34	0	Travel_Rare	419	Research & Development	7	4	Life Science	1	Female	53	3	3	Research Scientist	2	Single	11994	21293	0	No	11	3	3						
25	21	0	Travel_Rare	391	Research & Development	15	2	Life Science	3	Male	96	3	1	Research Scientist	4	Single	1232	19281	1	No	14	3	4						
26	34	1	Travel_Rare	699	Research & Development	6	1	Medical	2	Male	83	3	1	Research Scientist	1	Single	2960	17102	2	No	11	3	3						
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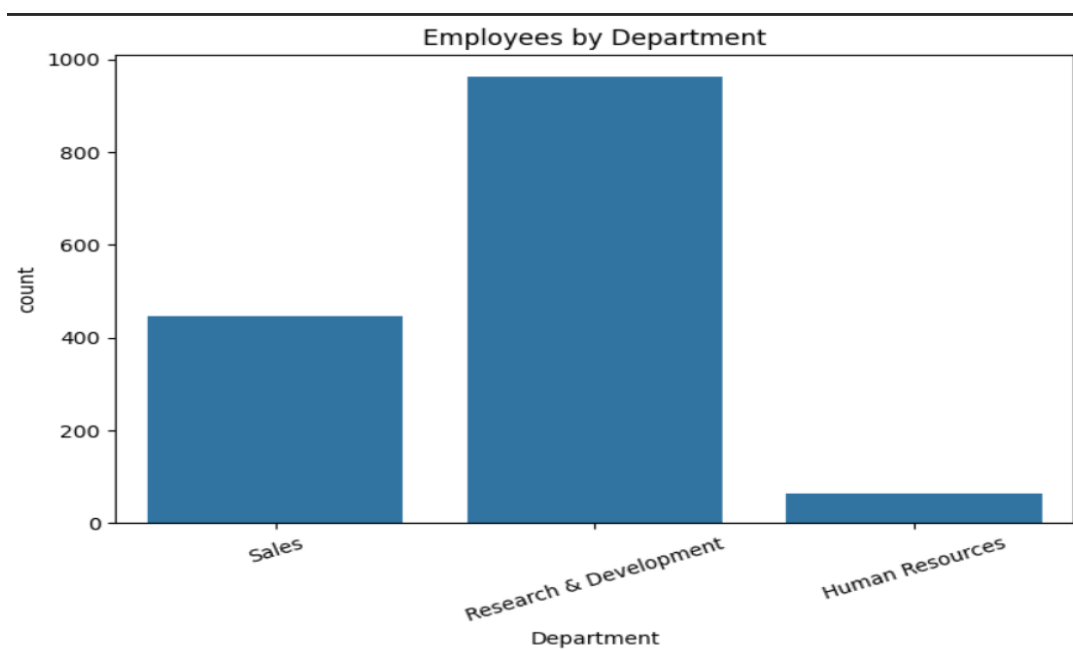
8. Exploratory Data Analysis

The attrition distribution shows the proportion of employees who remained with the organization compared with those who left. This provides an initial understanding of the employee retention pattern in the dataset.



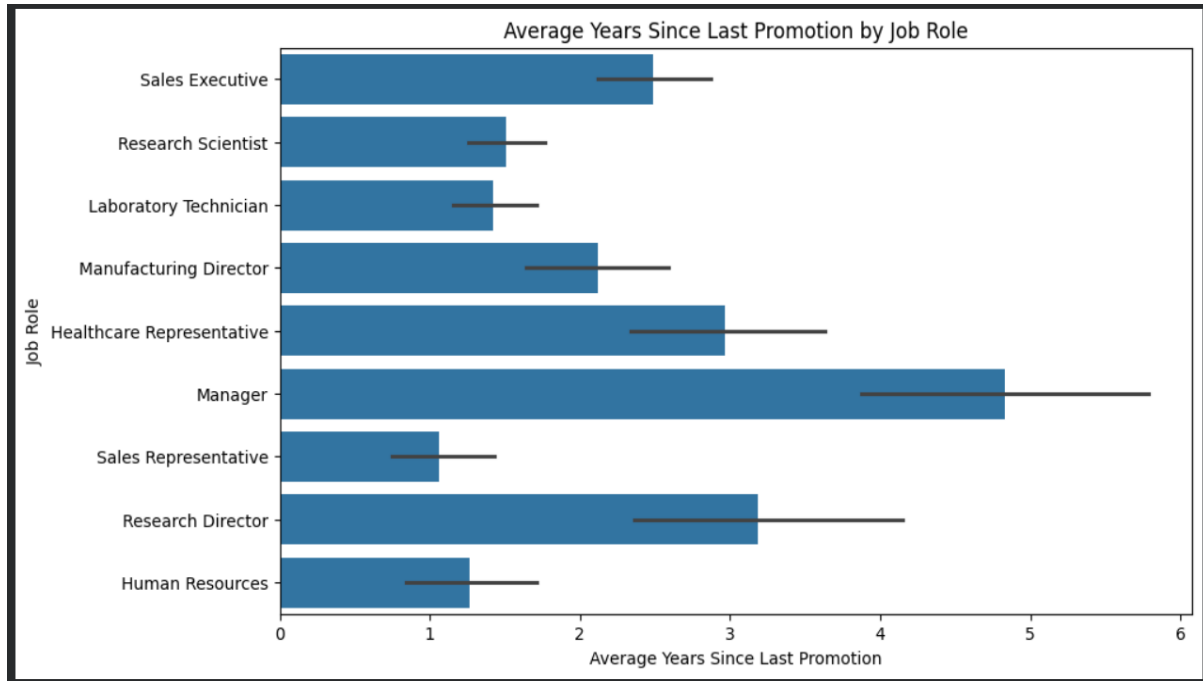
8.2 Department Analysis

Employee distribution and attrition patterns were examined across departments to determine whether career and retention patterns differ between organizational functions.



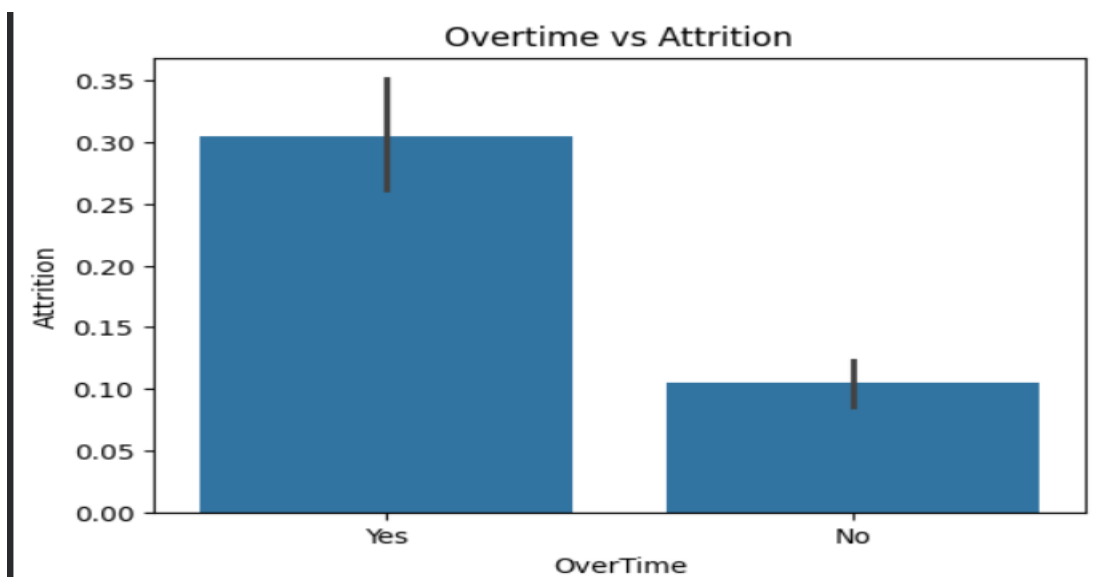
8.3 Promotion Gap Analysis

Years Since Last Promotion was analyzed across job roles to identify roles where employees may experience longer periods between promotions.



8.4 Overtime and Attrition

Overtime status was compared with attrition to understand whether working conditions show an observable relationship with employee retention.



9. Feature Engineering

9.1 Promotion Gap Ratio

Formula:-

$$\text{Promotion Gap Ratio} = \text{YearsSinceLastPromotion} \backslash \text{YearsAtCompany}$$

Explanation:

This metric represents the proportion of an employee's tenure at the organization that has elapsed since their most recent promotion.

9.2 Role Stagnation Index

Formula:-

$$\text{Role Stagnation Index} = \text{YearsInCurrentRole} / \text{YearsAtCompany}$$

Explanation:

This metric measures the proportion of an employee's company tenure spent in their current role. Higher values indicate a greater proportion of tenure spent in the same role.

9.3 Training Intensity

Formula:-

$$\text{Training Intensity} = \text{TrainingTimesLastYear} / \text{YearsAtCompany}$$

Explanation:

This metric provides a normalized view of training participation relative to employee tenure.

9.4 Manager Stability Ratio

Formula:-

$$\text{Manager Stability Ratio} = \text{YearsWithCurrManager} / \text{YearsAtCompany}$$

Explanation:

This metric measures the proportion of an employee's company tenure spent working with the current manager.

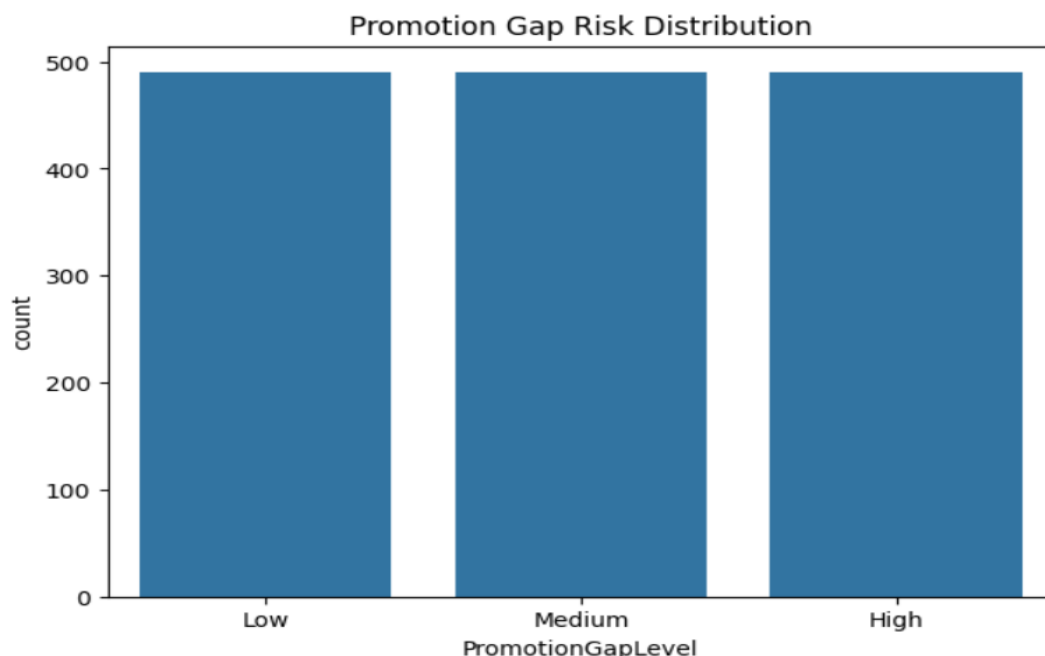
10.Career Path Clustering

K-Means clustering was applied to identify groups of employees with similar career progression characteristics. The clustering variables included company tenure, current-role tenure, time since last promotion, manager tenure, training participation, job level, and total working experience.

Before clustering, numerical variables were standardized using StandardScaler so that variables with different numerical ranges could contribute appropriately to the clustering process.

11. Promotion Gap Scoring

A Promotion Gap Score was developed to combine multiple career progression indicators into a single analytical measure. The score incorporates Years Since Last Promotion, Promotion Gap Ratio, and Role Stagnation Index.



12. Retention Opportunity Identification

Retention opportunities were identified by combining promotion stagnation indicators with employee retention-related characteristics. Employees classified as having a high promotion gap, who had not left the organization and demonstrated relatively positive job satisfaction, were flagged as potential retention opportunities.

13. Training Need Analysis

Employees with high promotion gaps and relatively low training participation were identified as potential training needs. This helps connect career stagnation with employee development opportunities.

14. Dashboard Development

Explain your Power BI dashboard.

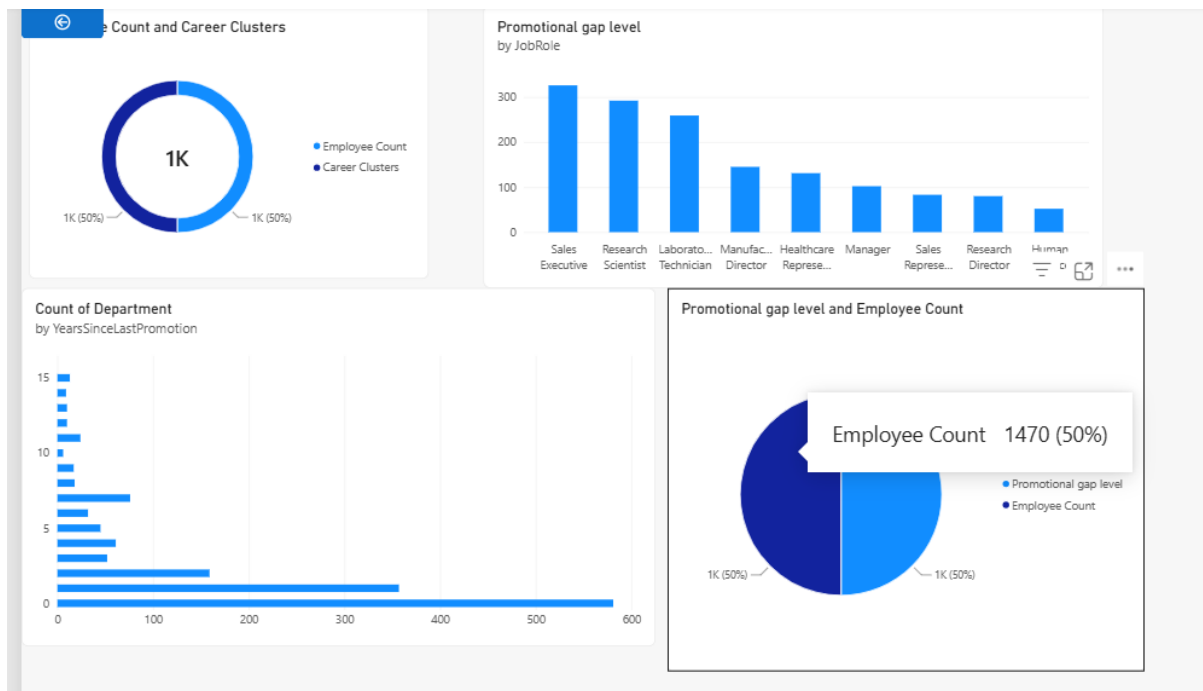
Power BI

Dashboard contains:

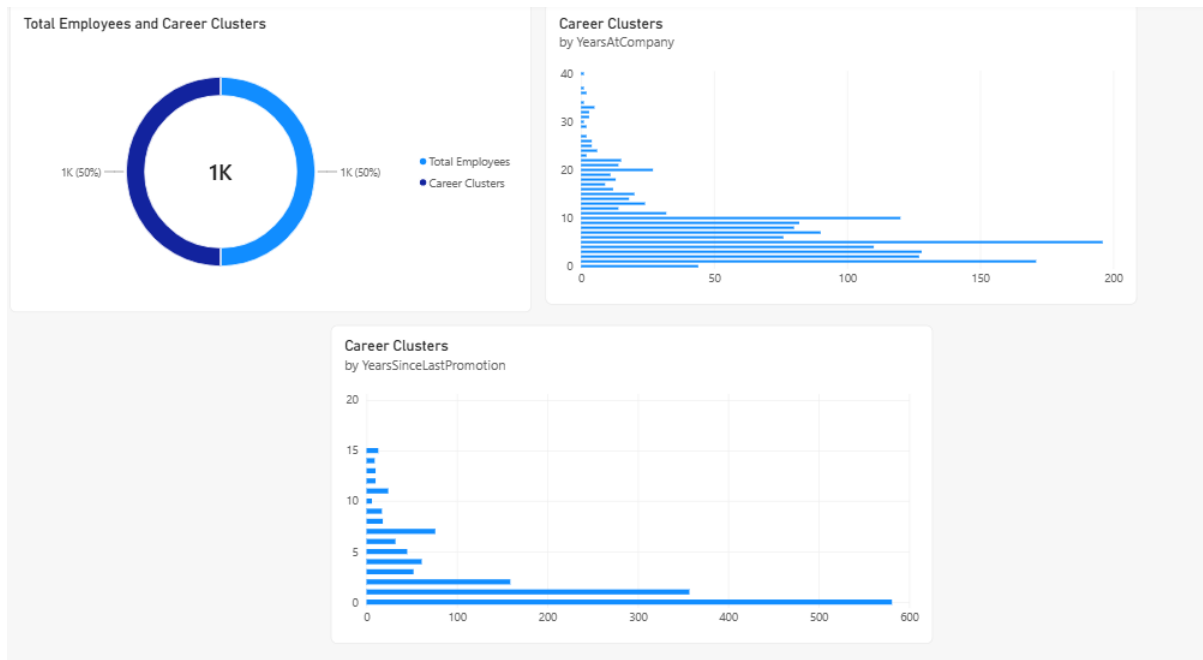
- Executive Overview
- Career Cluster Analysis
- Promotion Gap Monitor
- Retention Opportunity

Power BI was used to create an interactive business intelligence dashboard containing key performance indicators, career cluster distributions, promotion-gap analysis, department-level comparisons, and retention opportunity analysis.

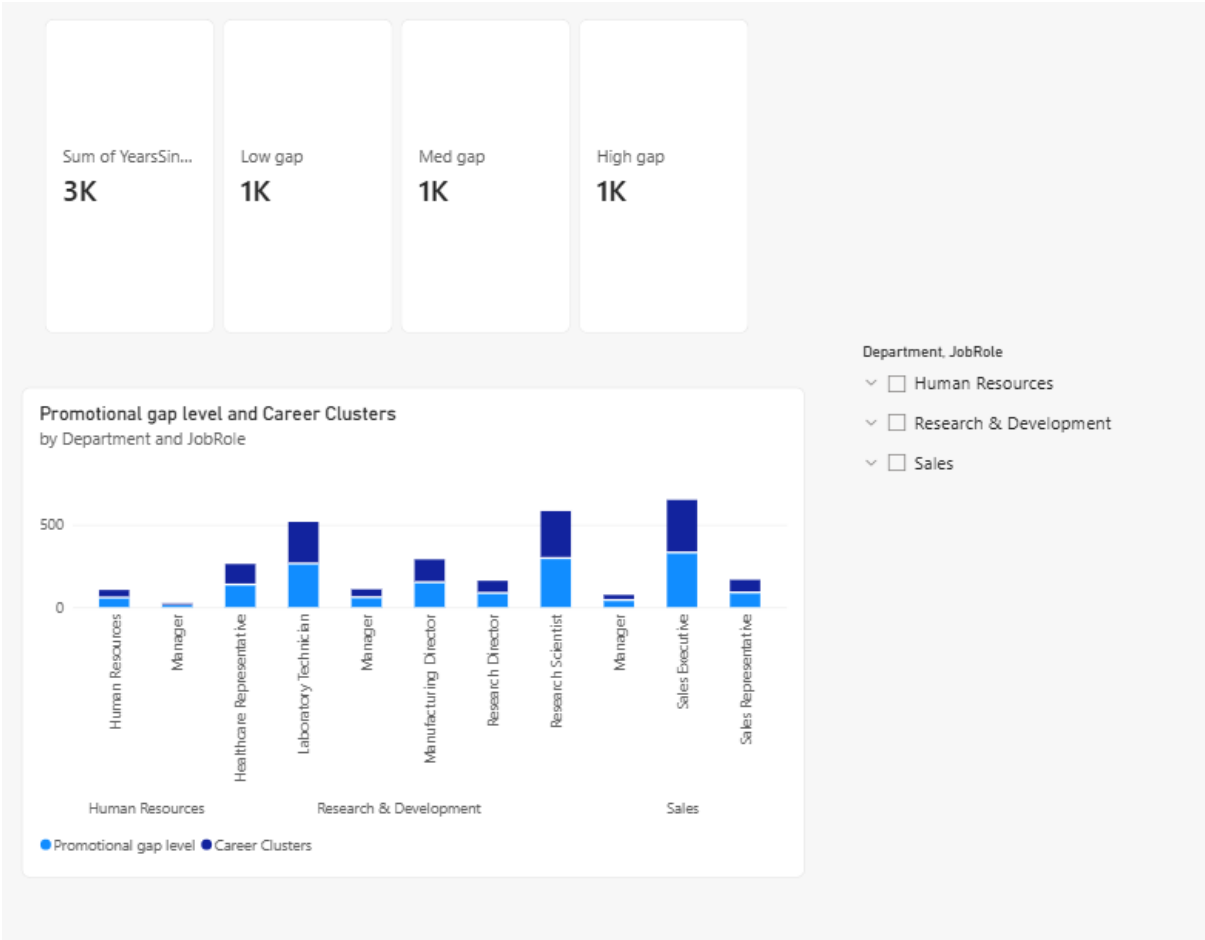
Executive Overview:



Career Cluster Analysis



Promotion Gap Monitor



Retention Opportunity

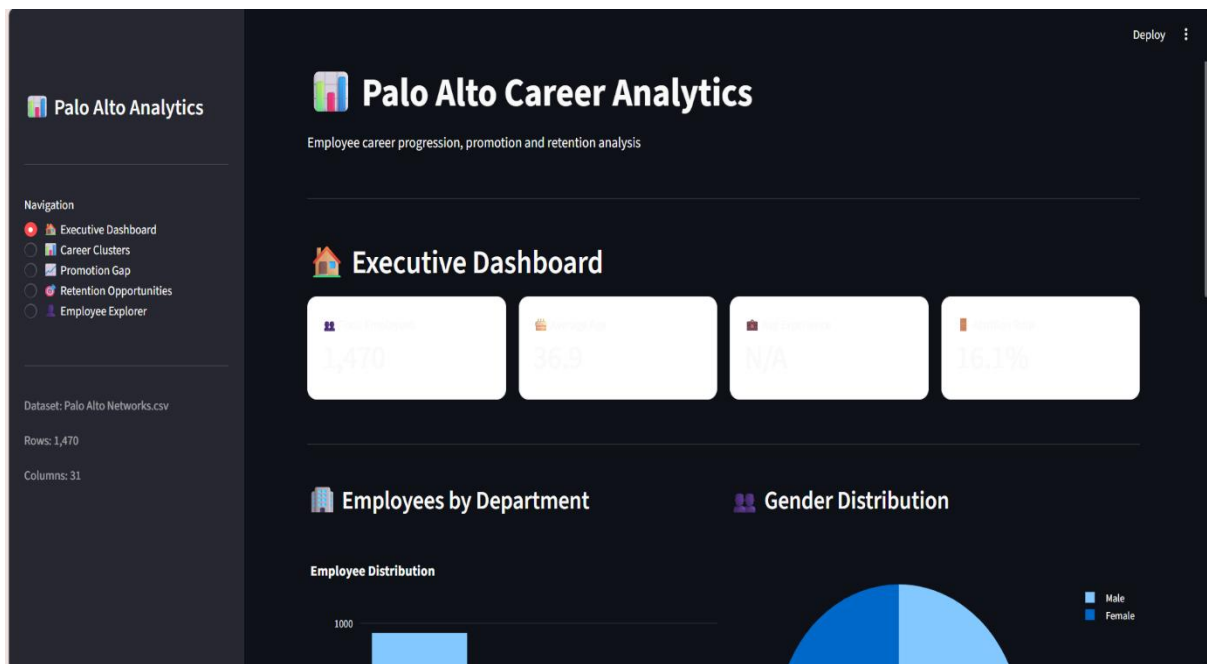
High gap
1K

Retention Opportunities
1K

Sum of Age	Department	Career Clusters	JobRole	Sum of YearsAtCompany	Sum of YearsInCurrentRole	Sum of YearsSinceLastPromotion	Sum of YearsWithCurrManager	Retention Opportunity
1846	Human Resources	52	Human Resources	277	163	66	165	
536	Human Resources	11	Manager	179	60	46	66	
5215	Research & Development	131	Healthcare Representative	1097	637	389	599	
8831	Research & Development	259	Laboratory Technician	1300	829	368	854	
2482	Research & Development	54	Manager	730	347	295	332	
5553	Research & Development	145	Manufacturing Director	1102	721	307	717	
3520	Research & Development	80	Research Director	875	503	255	486	
9997	Research & Development	292	Research Scientist	1493	956	440	937	
1752	Sales	37	Manager	563	251	152	242	
12026	Sales	326	Sales Executive	2444	1583	810	1525	
2520	Sales	83	Sales Representative	242	167	88	138	
54278		1470		10302	6217	3216	6061	

15. Streamlit Application

A Streamlit web application was developed to provide an interactive analytical interface. The application allows users to explore career clusters, promotion gaps, retention opportunities, and individual employee characteristics using interactive filters and visualizations.



16. Results

The analysis identified X employees in the high promotion-gap category. The clustering analysis produced X distinct career trajectory groups. X employees were identified as potential retention opportunities based on the defined criteria.

17. Business Recommendations

Based on the analysis,

Career Development

Employees with persistent promotion gaps can be considered for structured career-development discussions.

Training

Employees showing both promotion stagnation and low training participation can be considered for targeted learning programs.

Role Mobility

Employees with long periods in the same role may benefit from opportunities for expanded responsibilities or internal mobility.

Managerial Review

Career progression patterns can be reviewed at team or managerial levels to identify structural differences in progression opportunities.

18. Limitations

The analysis is based on the variables available in the dataset and therefore does not capture every factor that can influence career progression. Variables such as employee career aspirations, manager feedback, internal job availability, organizational restructuring, and individual performance context may not be represented.

The clustering results are exploratory and should be interpreted as analytical segments rather than definitive employee classifications. Similarly, the Promotion Gap Score is an analytical indicator developed for this project and should not be treated as an official HR performance or promotion decision.

19. Conclusion

This project demonstrates how employee career data can be analyzed to identify patterns of career progression and promotion stagnation. Through exploratory data analysis, feature engineering, clustering, and promotion-gap scoring, the project provides a structured approach to understanding employee career trajectories.

The resulting Power BI and Streamlit dashboards transform the analysis into interactive tools that can help HR stakeholders explore career clusters, promotion gaps, training needs, and potential retention opportunities.

Rather than focusing exclusively on predicting employee attrition, the project emphasizes understanding the career-related factors that may indicate opportunities for proactive employee development and retention.